

EXPERIMENT 6c – DATA VISUALIZATION USING POWER BI

Aim

To clean, preprocess, and transform the raw **Tourism Dataset**, load the cleaned data into Power BI, and build an interactive visual dashboard to analyze tourist locations, visitors, ratings, revenue, countries, and tourism categories.

Algorithm:

1. Import Dataset:

Load the **tourism_dataset (1).csv** into Power BI using **Get Data → Text/CSV**.

2. Deduplication:

Identify and remove duplicate rows across all columns in the tourism dataset.

3. Handle Missing Values:

Check the dataset for missing or null values in columns such as:

- Location
- Country
- Category
- Visitors
- Rating
- Revenue
- Accommodation_Available

Replace or remove missing values based on the column requirements.

4. Handle Data Types and Errors:

Ensure the correct data types are assigned:

- **Location** → Text
- **Country** → Text
- **Category** → Text
- **Visitors** → Whole Number
- **Rating** → Decimal Number
- **Revenue** → Decimal Number
- **Accommodation_Available** → Text/True-False

Remove or replace invalid values that may cause errors during data loading.

5. Transform and Prepare Data:

Clean and standardize the data by:

- Removing unnecessary blank rows.
- Correcting invalid values.
- Formatting text columns properly.
- Checking numerical columns for valid values.

6. Load Clean Data:

Load the cleaned tourism dataset into Power BI Desktop by clicking:

Home → Close & Apply

The cleaned dataset is then loaded into the Power BI data model.

7. Create Core Metrics:

Right-click the tourism dataset table and select **New Measure**.

Create the following measures:

Total Visitors

Total Visitors = SUM('tourism_dataset'[Visitors])

Average Rating

Average Rating = AVERAGE('tourism_dataset'[Rating])

Total Revenue

Total Revenue = SUM('tourism_dataset'[Revenue])

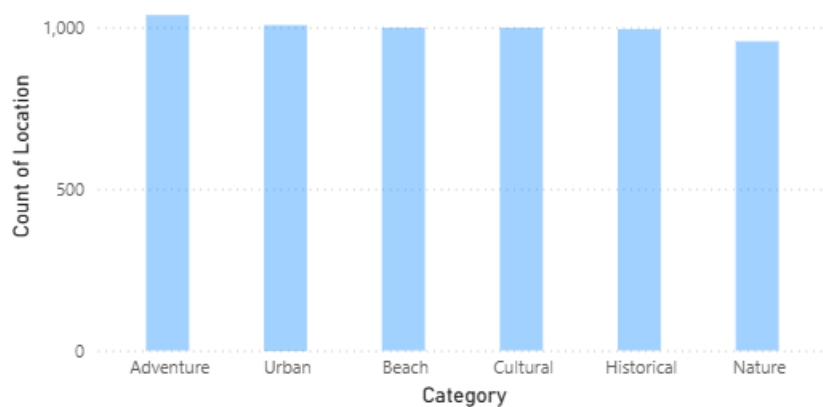
These measures can then be used to create KPI Cards and visualizations in your **Tourism Dashboard**.

OUTPUT:

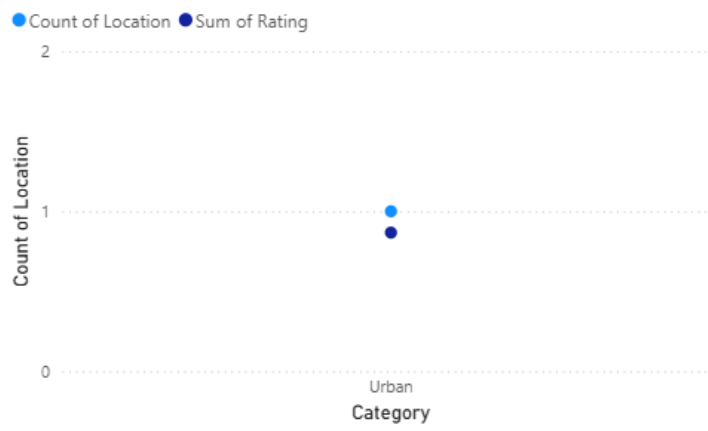
Category, Country

- ✓ ☐ Adventure
- ✓ ☐ Beach
- ✓ ☐ Cultural
- ✓ ☐ Historical
- ✓ ☐ Nature
- ✓ ☐ Urban

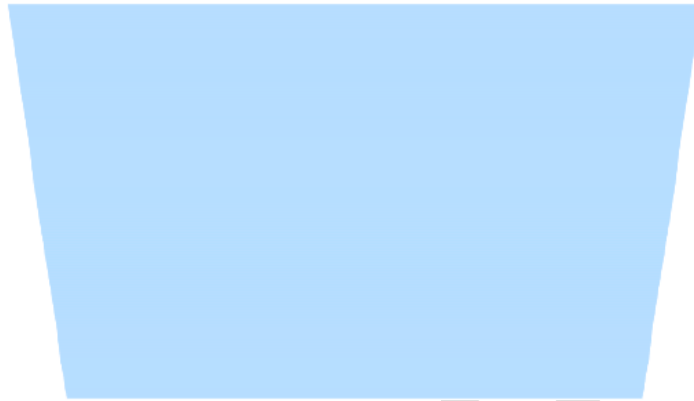
Count of Location by Category



Count of Location and Sum of Rating by Category



Sum of Revenue by Location



RESULT: